

Quiz 1 (SET-3)

1. a) $f(x,y) = \frac{x}{5} - \frac{y}{8} - 2 = 0$

in x intercept, $y=0$

$$\frac{x}{5} - \frac{0}{8} - 2 = 0$$

$$\Rightarrow \frac{x}{5} - 0 - 2 = 0$$

$$\Rightarrow \frac{x}{5} = 2$$

$$\Rightarrow x = 10$$

$\therefore \text{Start}(10,0)$

in y intercept, $x=0$

$$\frac{0}{5} - \frac{y}{8} - 2 = 0$$

$$\Rightarrow 0 - \frac{y}{8} - 2 = 0$$

$$\Rightarrow \frac{y}{8} = -2$$

$$\Rightarrow y = -16$$

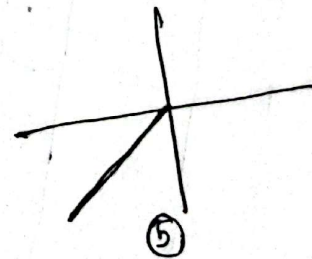
$\therefore \text{End}(0,-16)$

$$dy = -16 - 0 = -16$$

$$dx = 0 - 10 = -10$$

$$\therefore |dy| > |dx|$$

$\therefore \text{Zone} \rightarrow 5$



b)

Zone -5
Start (10,0)
End (0,-16)

convert to
Zone 0

Zone 0

$$x' = -y$$

$$y' = -x$$

$$(-y, -x)$$

Start (0, -10)
End (16, 0)

$$\therefore dy = 0 + 10 = 10$$

$$dx = 16 - 0 = 16$$

$$D = 2dy - dx = 2 \times 10 - 16 = 20 - 16 = 4$$

$$\therefore D = 4$$

$$\Delta NE = 2(dy - dx) = 2(10 - 16) = -12$$

$$\therefore \Delta NE = -12$$

$$\Delta E = 2dy = 2 \times 10 = 20$$

$$\therefore \Delta E = 20$$

Zone 0				Zone 5	
x'	y'	D	NE/E	$x' = -y'$	$y' = -x'$
0	-10	4	NE	10	0
1	-9	-8	E	9	-1
2	-9	-12	NE	9	-2
3	-8	0	E	8	-3
4	-8	20	NE	8	-4

pixels:

(10, 0), (9, -1), (9, -2), (8, -3), (8, -4)